CHAPTER 7



LIFE PROCESS: NUTRITION, TRANSPORTATION, RESPIRATION, EXCRETION

Some life processes like nutrition, transportation, respiration, excretion etc. are essential for the survival of a living organism. Energy is required for the processes to be carried out in our body. This energy is obtained from food.

- What do you think; do all organisms acquire energy from the same source?

We had studied earlier in food webs that plants are mainly the source of food for most organisms in the living world. We have also studied that solar energy is stored in the carbon compounds formed by the process of photosynthesis. Thus, plants are called as producers.

Consumers are dependent on producers in the whole living world. There are certain similarities as well as some fundamental differences in the life processes of producers andconsumers. In nutrition for example, while the first step in producers (mainly plants) starts from some simple substances like carbon dioxide, water etc. that in consumers(mainly animals), it starts with complex carbon compounds like carbohydrates, proteins etc. The life processes in multicellular plants and animals are carried out in specially arranged system of organs.

7.1 Development of organ systems

We had studied about the surface area to volume ratios of cells. We had found that as the size or diameter of a cell increases, its surface area to volume ratio decreases and so does the rate of functions going on within it.

- See the graph and suggest what do you think will happen if the length of a unicellular organism increases? (read values in graph on both axis per 10-16m)

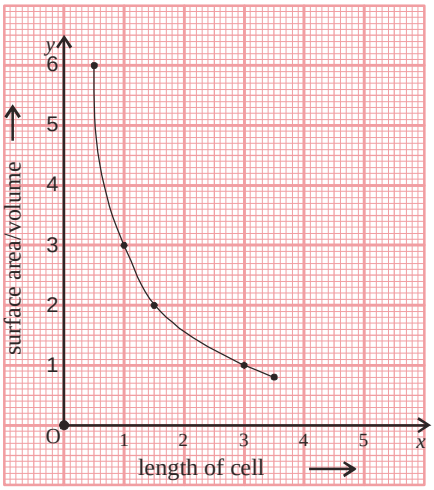


Figure-ure-1: Length of cell versus ratio of surface area to volume

You may have observed that as the length of a unicellular organism increases, its surface to volume ratio decreases. Apparently this could have a large impact on the availability of food resources for the organism. We may say that passage of resources into the body would be inadequate for the survival of the organism. With the increase in number of cells, the size of a multicellular body increases and this is just what happens in its body. Without a separate arrangement, resources fail to reach all the cells in its body. Moreover not all cells in the body of a multicellular organism are in direct contact with resources. Thus, there are

certain arrangements as follows such that resources reach all cells adequately all the time in a multicellular body -

- Enfolding in layers resulting in increase in area of absorption or passage of materials
- An arrangement by which several substances may be carried collectively via certain fluids thereby increasing rate of transport.
- Having mechanisms to reduce friction, likesecretion of mucous as required or having a moist surface etc.
- Favourable conditions for chemical reactions to take place.

Evolution has led to the development of such fundamental arrangements in the bodies of several multicellular organisms. Elaborate systems of membranes, reaction sites, tubes etc. are found in plants for photosynthesis, transport etc. while we have such elaborate systems in our digestive, respiratory, blood vascular, lymphatic, excretoryand other systems.

We shall study about all such systems and life processes in humans and in plants that are related to food and nutrition , starting from the intake of food, its utilization in the body and release of energy to excretion.

7.1.1 Digestive system in humans

Different components of food are required to maintain energy in our body. Processes like growth, repair and maintenance of body temperature require energy which is obtained by the breakdown of food components like carbohydrate, fats and proteins. We have studied and tested the presence of components of food in our previous classes(annexure to this chapter contains the details of the testing processes so that you may try doing them again). Apart from these, our body requires other substances like vitamins, minerals, salts, water etc. The process of digestion helps to maintain their required levels in our blood. This helps to fulfill the diverse requirement for resources of the numerous and diverse cells present in our body.Sugar (a type of carbohydrate) is required by most of the cells of our body. Sugar levels are thus maintained in our body. A fall in the level of sugar stimulatesthe transmission of a signal to the brain. A response from the brain results in contraction of muscles in the walls of the stomach, we feel it as a mild pain, signaling hunger pangs.

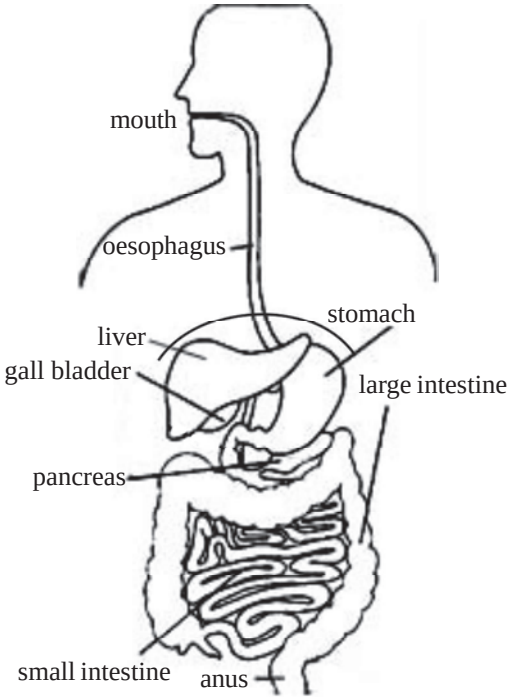


Figure-2: Human digestive system

- Think of some situations that make you feel hungry. What happens when the scent of a favorite dish reaches you?

Our digestive system mainly consists of a tube (nearly 27 feet long) extending from mouth to anus(also called as the digestive tract or gut). Different regions of this tube have different names due

to structural and functional variations in them. The walls of this tube are folded in layers, somewhere the folds are numerous, long and dense as in the intestine somewhere it'sotherwise as in the tube between mouth and stomach. In some areas within the walls of the tube or outside it are certain organs that secrete substances that aid in digestion, these are called as digestive glands like salivary glands in mouth and gastric glands in stomach are internal while liver, gallbladder and pancreas are external. Secretion from these glands as well as from certain other cells of the wall of the digestive tube helps in the process of digestion. Large molecules of our food are change to smaller molecules in this process.

- What are the different parts that have been shown in the diagram of human digestive system?

7.1.2 The process of digestion

Digestion starts as soon as we start chewing food and saliva mixes with it. Saliva is secreted by numerous salivary glands in our mouth.3 pairs of them are quite large.Saliva has an effect on certain carbohydrates like starch in the food that we eat. Let us do an activity to study this.

Activity-1

We would need a teaspoonful of wheat flour (you could dilute and use the starchy fluid drained out of boiled rice as well), distilled water, dropper and iodine solution (preferably 1% solution of tincture iodine), two beakers, two test tubes and a test tube stand.

Now, add half a teaspoon of wheat flour to half a beaker (around 125ml) of distilled water, mix well to prepare a solution. Now take a little of this solution in a test tube add a few drops of tincture iodine and see if it turns blue black or not that is gives confirmatory test for presence of starch . In case it doesn't, add more of wheat flour till you get a confirmatory test for starch. Now start your experiment with this solution. Now take two test tubes and label them as A and B respectively. Keep them in a test tube stand. Add 20-25 drops of wheat flour solution to both test tubes.

Now to test tube A add a similar amount of your saliva. Do not add anything to test tube B. Leave the set ups for about two hours. Then add a few drops of iodine solution to both test tubes A and B. Observe which one shows presence of starch.

- What happened after adding iodine to both test-tubes A and B?
- Why do you think this may have happened?
- What do you think is the effect of saliva on wheat flour solution?

Now let us observe the nature of the medium (that is acidic, basic or neutral) on the effect of saliva.

Activity-2

As in activity 1, use wheat four solution or starch solution from drained from boiled rice for experiment. Now take three test tubes and label them as A, B and C respectively. Add 8-10 drops of lemon juice in test tube A. Add around the same number of drops of washing soda solution (prepared by adding a pinch of soda to a test tube full of distilled water) to test tube B. Add same number of drops of distilled water to test tube C. Use universal litmus paper to find out the pH of each of the

substances of test tube A, B and C respectively. Now add 20-25 drops of wheat flour solution to each of the test tubes. Now add a similar volume of saliva to each test tube and leave for around 2 hours. After two hours, use 1% iodine solution to test the presence of starch in each of the test tubes.

- What is the nature of solution (acidic, basic or neutral) in test tubes A, B and C respectively?
- Which of the solutions A, B and C shows the presence of starch?
- Why do you think that even though we started with the same amount of solution of wheat flour and saliva, our experiment does not show presence of starch in all the test tubes?

In this activity we observe that in test tubes B and C there in an effect of saliva on starch while in A we do not see any effect indicated by presence of starch till the end.

A substance present in saliva called as 'enzyme' (in this case salivary enzyme) is found to act only in specific conditions like particular pH range (here basic and neutral), temperature etc.

Do you know?

Stomach with a window and discovery of effect of enzyme on digestion

In 1822, an American soldier named Martin was accidentally shot in the stomach and was brought to Dr. Beaumont. The doctor treated his wound which healed with a hole in the stomach covered with a flap of skin. A pipe could be inserted through this hole and digestive juice could be drawn out from the stomach. Beaumont started experimenting with this juice and found that food was digested inside the stomach at a much faster rate as compared to that outside the body using the same digestive juice. Thus he recognized that certain specific conditions were required for the digestive juice to function in the process of digestion which he concluded was not only a mechanical process of grinding and churning but also a chemical process. Around a decade after this finding, a French scientist, Anselme Payne discovered a substance extracts of barley sprouts that could change starch to sugars, he named it as diastase. It was later found to be a group of compounds called as amylase which is found in parts of the human digestive system (salivary glands, pancreas etc.), several bacteria, some types of fungi, certain plants etc. These substances in the digestive juice, in barley sprouts and other sources were collectively named as 'enzymes' in the year 1877. The word meant 'to leaven'.

Most enzymes are proteins. There are numerous of these present within a single cell of our body. All reactions related to synthesis and breakdown are speeded up and mediated by enzymes. Enzymes have some remarkable properties. Take for example the enzyme 'Pepsin' secreted in our stomach, it can digest around 50 times its weight of proteins in our food in an hour and continue doing so for some hours. Most of our life processes are accomplished by enzymes secreted within cells or by cells of our body. Digestive enzymes are either secreted directly from walls of gut into the cavity or by glands outside the gut and flow into it by some other tubes.

7.1.3 Function of the digestive system

After digestion in the mouth, food reaches the stomach. The passage of food is lubricated by mucus secreted by walls of the gut and pushed forward by a process called as 'peristalses. This is a series of wave like motion due to alternate contraction and relaxation of muscles in the wall of the

gut. Food passes from mouth to 'oesophagus' (literally meaning to eat and carry) via a muscular structure called pharynx.Thereafter by peristalsis it passes through oesophagus to the stomach.Hydrochloric acid and an enzyme called pepsin (which functions only in acidic medium) are mainly released from glands in the wall of the stomach that help in the digestion of proteins. Churning, grinding and mixing with digestive juices in the stomach turns food to a thick fluid that passes in spurts into the small intestine. Small intestine is nearly 20 feet long.The process of digestion completes here and nutrients are absorbed into vessels (blood and lymph) that carry them to other parts of the body. Digestive juices from liver and pancreas that reach the small intestine aid in the process of digestion. Pancreatic juice contains sodium bicarbonate that renders the medium alkaline and enzymes present in the digestive juices become functional. Thus most of the carbohydrates, fats and proteins are digested here. There are certain stimulators as well that are secreted from different parts of our digestive system (these are certain hormones). You would be surprised to know that there are several colonies of microbes as well present in our gut that help in the process of digestion.

Do you know?

Amylase, trypsin and lipase present in pancreatic juice help in digestion of carbohydrates, proteins and fats respectively. Constituents of food are broken down from larger and complex to small and simple compounds. Carbohydrates like starch are digested to sugars like glucose, proteins to amino acids and fats to fatty acids and glycerol. Bile present in bile juice secreted by liver has certain salts that help to convert fats to small droplets or micelles (a process called as emulsification) aiding in the functioning of lipase.

The numerous foldings of the small intestine are also called as villi. The surface area of absorption increases due to the villi. The blood vessels and lymph vessels over and inside these villi help to carry the absorbed digested nutrients to different parts of the body. Some water and salts are also absorbed in the small intestine. The rest of the food which is largely undigested matter passes to the large intestine where most of the water and certain salts are absorbed. The stool (mainly undigested matter) and certain gases pass out of the anus.

Do you know?

A wonderful ecosystem specific to an individual exists in our gut. There are several colonies of bacteria in it. The life processes of these bacteria not only aid in digestion they also convert certain proteins and vitamins into forms usable and absorbable in our body. Several gases like hydrogen sulphide, methane, ammonia etc. are formed due to the life processes of these microbes.

- How does digested food reach different parts of our body?

7.1.4 Structures related to transport and their function



William Harvey (1578-1657)

The structures involved in transport of digested food and several other substances are mainly vessels carrying the fluids blood and lymph. The whole network is called as blood vascular system. People even a thousand year ago knew that blood flows in vessels that are connected to the heart. They believed that with each beat blood flowed out of the heart into the vessels to different parts of the body and was

absorbed back into the heart. Different blood vessels were observed and named but the actual mechanism of functioning of the heart and the flow of blood was not known until the researches done from 15th to 17th century. The credit goes to a doctor of the 17th century, William Harvey who described the mechanism of flow of blood in our body and the role of heart. Harvey was inspired by the discovery of valves in veins allowing flow of blood only in a certain direction. Harvey performed several experiments to observe the flow of blood. You may also try doing one of his experiments.

Activity-3

Tie the hand of a friend as shown in figure 3 B. Ask your friend to close his fist tightly and quickly move the hand from elbow to fist at least 4-5 times. Now keep your fingers as shown in the figure on the bluish prominent vessels of your friend's hand. Press your fingers down and move one towards fist and the other towards elbow.

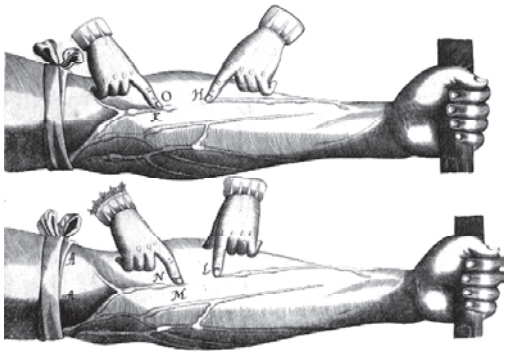


Figure-3: Illustration of an experiment by William Harvey

- Did the bluish colour disappear between these points?

Now remove your finger towards the fist of your friend. Observe what happens. Now repeat the experiment and remove your finger towards the elbow of your friend.

- Towards which side (elbow or fist) do you find blood vessels fill up faster? This shows the direction of blood flow.

Harvey observed that blood flows from palm to elbow (as you may have also observed), thus these vessels were definitely carrying blood from our body to the heart. Thus he defined veins to be vessels that carried blood from different parts of the body to heart and arteries as vessels that carried blood from heart to different parts of our body. He suggested that blood flowed in a circular pathway in our body starting from heart and returning to it. Harvey could not show the connections between veins and arteries but suggested their presence. Their presence was later confirmed when they were observed and named as capillaries.

Veins can be observed in different parts of our body as apparently bluish vessels under the skin. Blood flowing through them is dark red and not blue. Arteries are usually present deeper inside so we may hardly observe them. Some of them are present near the surface like those in our wrist, ankle or side of neck. We can feel them pulsating with each beat of the heart. Blood flowing through them is bright red in colour. Let us feel one of the arteries.

Activity-4

Keep thefingers of one hand on the wrist of the other as shown in figure 4.

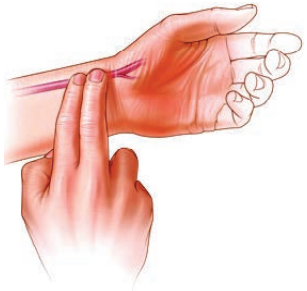


Figure-4

- Could you feel it pulsating?

As the heat beats, so does the arteries and we feel the beat as a pulse in an artery.

Do you know?

All blood vessels are made up of cells. The space between cells increase or decrease according to conditions around them as substances flow in or out of the vessels. For example if you have a cut, certain substances flow out that help in healing and forming the clot.

7.1.5 The role of heart in transport

Heart functions as a pump in our body as well as that of most other animals. Around 7000 liters of blood (4.7-5.5 liters in each of the nearly 1500 circuits) is pumped through our heart which is no more than the size of our fist. There are 4 major chambers in our heart. The two upper chambers are called as auricles and the lower ones are called as ventricles. There are valves between auricles and ventricles as well as between auricles and blood vessels or ventricles and blood vessels that arrest backflow of blood. The left side of the heart is separated from the right by a partition called septum thus no mixing of blood of chambers in right with those in left occur. The chambers in the right have deoxygenated blood while those on left have oxygenated blood. Auricles contract together and blood flows out through them into the ventricles. As ventricles contract blood flows out from the heart into arteries and simultaneously, blood flows into the auricles from veins.

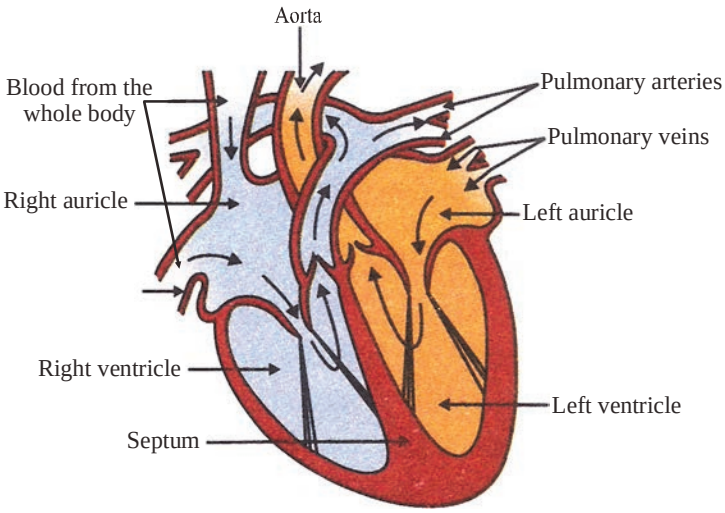


Figure-5A: Chambers of Heart

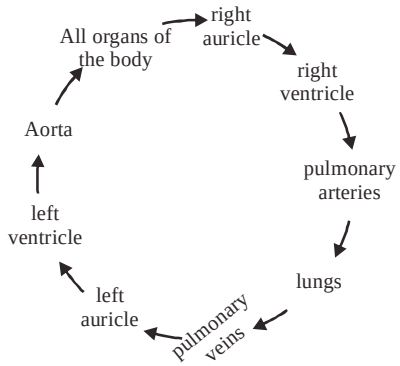
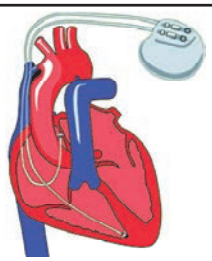


Figure-5B: Circuit of blood flow

The function of the heart is controlled and coordinated via nerves from the brain as well as an elaborate network of nerves in the heart.

Do you know?

We call the major network of nerves of the heart as Sino auricular node. When this fails to function, a pacemaker device is implanted into the heart that carries out the function of this node for some years.



Pacemaker

The circuit that we see in fig 5B actually represents two pathways that differ functionally. That is in one pathway de-oxygenated blood from heart is oxygenated via lungs and returned to it. While in the other oxygenated blood is deoxygenated via all parts of the body (including thelungs) and returned to heart. The two pathways are shown in Figure- 6. We can trace these pathways from the heart. As the left ventricle contracts oxygenated blood flows out into the large artery called aorta that divides into several arteries carrying blood to the tissues. Capillaries thereafter carry blood to the cells and carry deoxygenated blood back to veins that collectively carry back blood to the right auricle of heart. This is one pathway. The other starts with contraction of right ventricle as deoxygenated blood flows out through pulmonary arteries to the lungs. There arteries and capillaries carry blood to sites of gaseous exchange from where oxygenated blood is returned via capillaries to veins to the pulmonary vein that carry blood to the left auricle of heart.

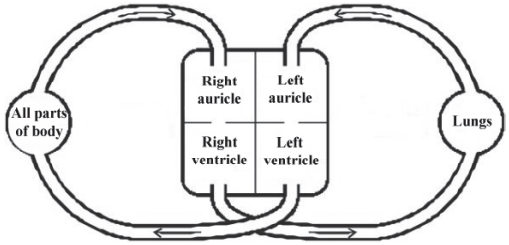


Figure-6: Double circulation of blood through heart

- Does oxygenated or deoxygenated blood usually flows through our veins?
- "Vein do not always carry deoxygenated blood" do you agree to this statement? Why/ why not?

The contraction of the ventricles creates a pressure that maintains blood flow in the arteries. There is hardly any pressure in the veins. We had studied that the valves in veins arrest backflow of blood. These along with contraction and relaxation of muscles around veins maintain blood flow in them.

Do you know?

The pressures on the walls of the ventricles when they contract and when they relax are represented by the values 120 and 80 respectively as normal blood pressure. The sounds of heart beat are sounds of closing of valves between auricles and ventricles and those between the ventricles and blood vessels as the ventricles contract and relax respectively.

- From where does blood flow into the right auricle?
- Where does blood go from right ventricle?
- Where is blood oxygenated?

The division in aorta is such that the network formed carries blood nearly at the same rate to all parts of our body as it flows out from the heart. We call this a parallel system of flow as the arteries reach the organs simultaneously. A series arrangement of arteries enhances resistance and slows down the rate of flow as blood passes from one part to another. This facilitates the exchange of materials between blood and cells.

- What do you think would happen if blood flowed in series from heart to other organs of our body?

The flow of blood from intestine to liver and to heart is a series pathway. It is inevitable that there should be nutrients in the blood flowing into the liver. There are several such nutrients the higher levels of which would be fatal for the body. Some such substances like excess sugar and fat

are removed from the blood in the liver. When their level falls they are returned back to blood. Several toxic substances are also either removed or changed into non- toxic substances in the liver. Most of these functions occur between cells of the liver and blood.

7.1.6 Role of blood in transport

Our knowledge of blood and its constituents is nearly 400 years old. After the discovery of cells, an observation of blood showed that it was not just a watery fluid but it contained several types of cells as well. Blood was thus said to be composed of two major components-

1. A fluid part that was called as plasma 2. Certain structures found in plasma that were called as blood cells.

Blood cells were later categorized as red blood cells, white blood cells and platelets. The red blood cells were found to be non- nucleated disc shaped structures having an iron rich protein compound called as hemoglobin. This is a pigment that rendered the cells red. Oxygen binds with the iron of hemoglobin and nearly 97% Oxygen in our body is carried in this way. Oxygen flows from blood to cells where its concentration in less and carbon dioxide reaches blood from these cells. The white blood cells lack colour and are of different shapes and sizes. Our body is protected from several infections and diseases due to these cells. A clot is formed in the area of a cut or injury by mainly another group of cells called as platelets. There are several other substances in blood that have different functions for example a group of substances circulating in blood dissolve unnecessary clots. Glucose, calcium, potassium, hormones, enzymes, urea, salts are among several other components of blood. The levels of several components of blood are indicators of our health and can be tested.

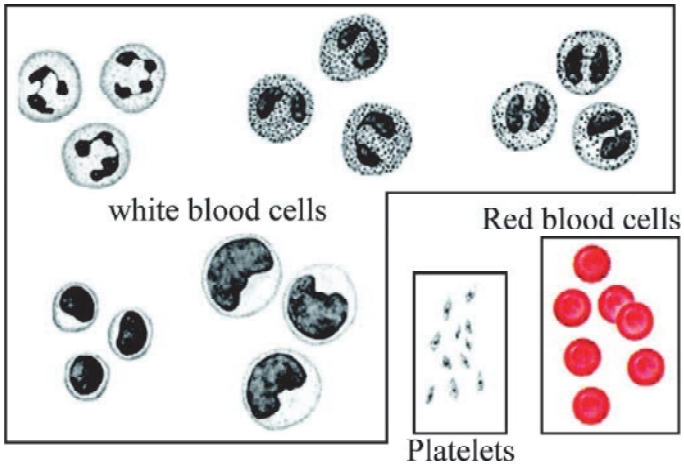


Figure-7: Some types of blood cells

Do you know?

You may have heard about cholesterol (a type of fat). Accumulation of this in the arteries of heart can cause a heart attack. A particular amount of this is essential for the lubrication between cells especially in our gut. The absorption and use of certain vitamins also are facilitated by cholesterol. Thus a certain amount of it is not harmful but rather very useful for our body.

7.1.7 The role of lymphatic system in transport

Lymphatic system is mainly a large and complex system of vessels,several nodes that are tissue aggregates and certain organs. It is believed that the pressure in blood vessels cause some plasma to drain out of blood vessels. This combines with intercellular fluid. This is lymph which is collected into vessels (like the capillaries and veins) that help to return the drained out fluid into veins.The flow of lymph is maintained by peristalsis and valves in the walls of lymph vessels. Lymph

contains some proteins, fats,glucose,salts etc. and cells called as lymphocytes which are specialized white blood cells. Protection from infections, removal of toxic substances, repair of tissues etc. are some functions of this system. Fats are mainly transported through lymph vessels. Lymph does not carry gasses which are transported by blood.

7.1.8 Respiration and associated structures

We find that blood and lymph carry nutrients to our cells where food is oxydised and energy is released, a process called as respiration. This energy helps to run all life processes. All living organisms whether plants or animals respire all through the day. This requires a constant exchange of gases between the body of the organism and its environment. There is an elaborate system of vessels, stomata and other structures in plants while there are vessels, gills or skin or lungs and other structures in animals that facilitate gaseous exchange. We have a pair of lungs and respiratory tubes extending from nose to lungs that help in gaseous exchange, the whole network being termed as respiratory tract (see Figure- 8). Right lung is slightly larger than the left one.

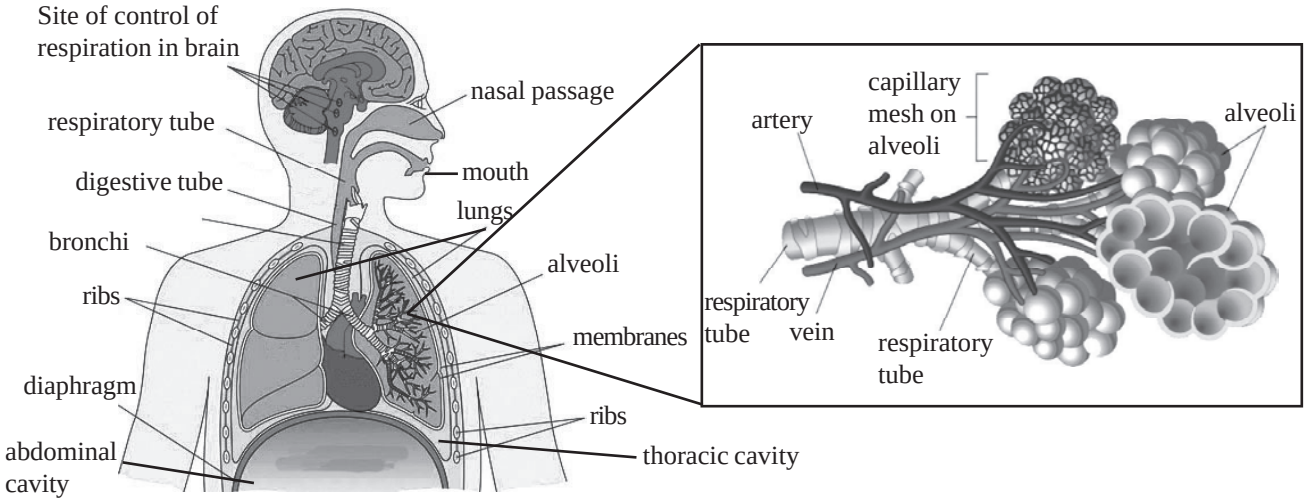


Figure-8: Respiratory System

Lungs are spongy structures with a vast branched network of respiratorty tubes. The extremities of the tubes have bunches of several air sacs which are bubble like structures called as alveoli. If all the alveoli were spread out they would cover an area of nearly 100 square meters. This providesa vast surface for gaseous exchange. Blood vessels and alveolar walls have a fine interface facilitating oxygenation of blood that has come from the heart.While mainly oxygen diffuses into the blood vessels, carbondioxide diffuses out from blood vessels to alveoli.The lungs may be affected by infections, dust and smoke.

Breathing in or out is a process that ultimately helps to maintain the internal environment of cells by maintaining the level of certain substances in them. The movement of diaphragm which is a muscular structure between thoracic and abdominal cavity and movement of ribs by muscular movements facilitates expansion or contraction of lungs. When diaphragm contracts and becomes flat, and rib cage extends out,volume of thoracic cavity increases and air flows into lungs through

tubes starting from our nose. Contrary to this, relaxation of diaphragm causes most of the air to flow out of the lungs. A maximum of nearly 4-6liters of air may fill into the lungs with each breath. Not all of this flows out in a living individual. Nearly 1.5-2.5 liters of air is always present in the lungs.

If one of our lungs become non functional, the other takes over the function. This happens in the case of our kidneys as well. We may never feel any difference all through our life. Several functions go on in our body that complement each other. All systems of our body work in mutual coordination. Certain structures show such pathways. One such structure between respitratory tract and gut is shown in Figure-8.

Warning!

Smoking causes swelling in the respiratory tract. It also stimulates secretion of more mucus that eventually narrows the passage. Incidence of infections also increase thereby leading to fatal diseases the symptoms of which start with coughing, difficulty in breating etc. Smoking adversely affects the individual an its family.

- Have you ever felt the connection? How?

While swallowing food our breathing stops for a moment. Lets do an activity to feel this.

Activity-5

Hold a small piece of paper near your nose and observe it as you breathe in or breathe out.The paper sticks to your nose when you breathe in while moves away from nose as you breathe out. When you stop breathing the paper does not move.

Now try to swallow food.

- What do you observe? Why do you think this happens?
- Why do you think we get sometimes get choked while swallowing food?

The valve between digestive and respiratory tract remains open while we breathe. When we swallow food it stuts close the respiratory tract. While swallowing hurriedly it may not close properly and food may pass into our respiratory tract. This instantly stimulates muscle contraction in it to expel food from it as we start coughing. Air filled in the tract is pushed out along with the food particle.

While breating air passes in and out of the respiratory tract. The air we breathe out or 'exhale' is slightly warmer,with less oxygen and more carbondioxide and more water vapour compared to air we breathe in or 'inhale'. Let us do an activity to compare cabondioxide levels in atmospheric air(we inhale) and air exhaled to observe the difference.

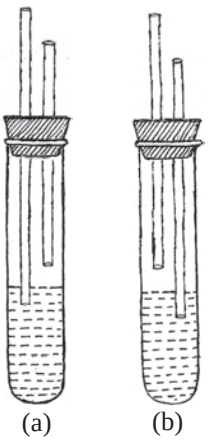


Figure-9

Activity-6

You would need two boiling tubes/test tubes, a test tube stand, lime water (freshly prepared), two straws/empty refill (nib removed)/two glass tubes, rubber corks with two holes, a little wheat dough/plasticine, watch. Arrange your apparatus as shown in practical section. You may use a syringe/pichkari in place of sucking air out. Pour same volume of freshly prepared lime water in both boiling tubes and

mark them as A and B respectively. Blow air into 'A' through a straw while use a syringe or pichkari to pass air into 'B'.

Do you know?

At resting phase, nearly a volume of 1 square foot of oxygen is used in our body. Nearly ten times of this volume is used during exercise. Carbon dioxide formed and expelled is slightly less than oxygen used by the body. If the amount of carbon dioxide increases and that of oxygen decreases in blood we start panting. When that of oxygen increases and carbon dioxide decreases we start breathing normally.

- Lime water turns milky faster in which of the two boiling tubes? Why?

Our observation that lime water turns milky faster when we blow air into it indicates that there is more CO_2 in exhaled air than in inhaled air(or atmospheric air)..

- Where did this extra CO_2 in exhaled air come from?

We would have to know about the reactions in our cells that ultimately lead to this observation. We have studied that digested food(carbohydrates especially sugars, proteins mainly amino acids and fats or fatty acids)and gases reach cells via blood. Energy is obtained from all the consituents of digested food. Let us take glucose for example (see fig 9), oxidation of glucose releases a large amount of energy. When we exercise or walk a lot, there is a dearth of oxygen in the cells of our muscles when glucose still breaks down and energy is released. The amount of energy is lesses as that during complete oxidation. There are certain chemicals in which energy is stored and utilized as and when required, these are called energy currencies and a compound named as adinosine tri phosphate(ATP) is such a compound.

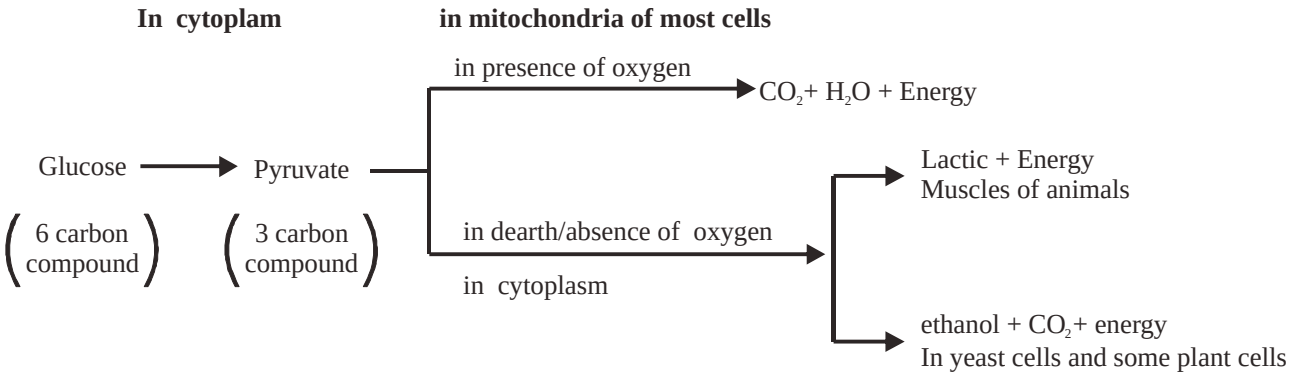


Figure-10: Pathways of breakdown of glucose

Do you know?

Fermentation is a process of breakdown of glucose in the absence of oxygen(also called as anaerobic condition). This naturally occuring process has widespread industrial use and several carbon compounds like ethanol, lactic acid,citric acid etc. are produced.

The process of breakdown of glucose to release energy occurs in the whole living world by pathways as shown in Figure- 9. Respiration is a process that involves all the processes from gaseous exchange to the breakdown of digested food to release energy. Several products of respiration are useful to our body while some are harmful and are expelled from the body.

- Which product of respiration (after glucose breakdown)is expelled from our body? How?
- Apart from glucose what other compounds release energy?
- What are the other processesin which products formed are expelled from our body?

Certain products formed by our life processes or those that we consume and are in excess are removed from our body. For example mainly carbon dioxide is removed as we breathe out, excess salts are mainly removed in sweating, urea and uric acid is mainly removed by urine.

We have a system that mainly forms urine from blood that flows through it, controls its volume and expels it from our body which we call as excretory system.

7.1.9 Structures and functions related to the excretory system

Normally about 1.5 litres of urine is produced daily. You may have observed that during hot and dry seasons the volume reduces.

- What happens to the volume of unine when we drink a lot of water?
- Does the volume of urine increase if we sweat a lot?

When we drink a lot of water the amount of urine expelled increases. Apart from water, urea and uric acid there are several other substances present in our urine. Doctors usually ask us to get our blood and urine tested to compare the levels of these sustances that indicate several aspects of our health.

Activity-7

Let us study some values of levels of certain substances in the urine and blood of a person to observe how they vary. A table has been prepared by collecting some values from blood and urine reports of a person and presented in this section.

Table-1: Some values for comparative study of substances in blood and urine

Substance	In Bood		In Urine	
Result	Normal range	Result	Normal range	
Glucose	82mg/dl*	70-100mg/dl	65mg/dl	50-80mg/dl
Urea	29mg/dl	15-40mg/dl	35mg/day	20-30mg/day
Uric acid	7.5mg/dl	3-5mg/dl	800mg/day	600mg/day
Total protein	7.2g/dl	6-7.5g/dl	0.9g/day	<0.1g/day

*Value shows fasting sugar(in result and range)

mg/dl milligram/decilitre

g/dl gram/decilitre

- Observe the table and find out the substances that are have value above the normal range in urine.
- What are the substances in blood that have values above normal range?
- Do same substanceshave high values in both blood and urine?

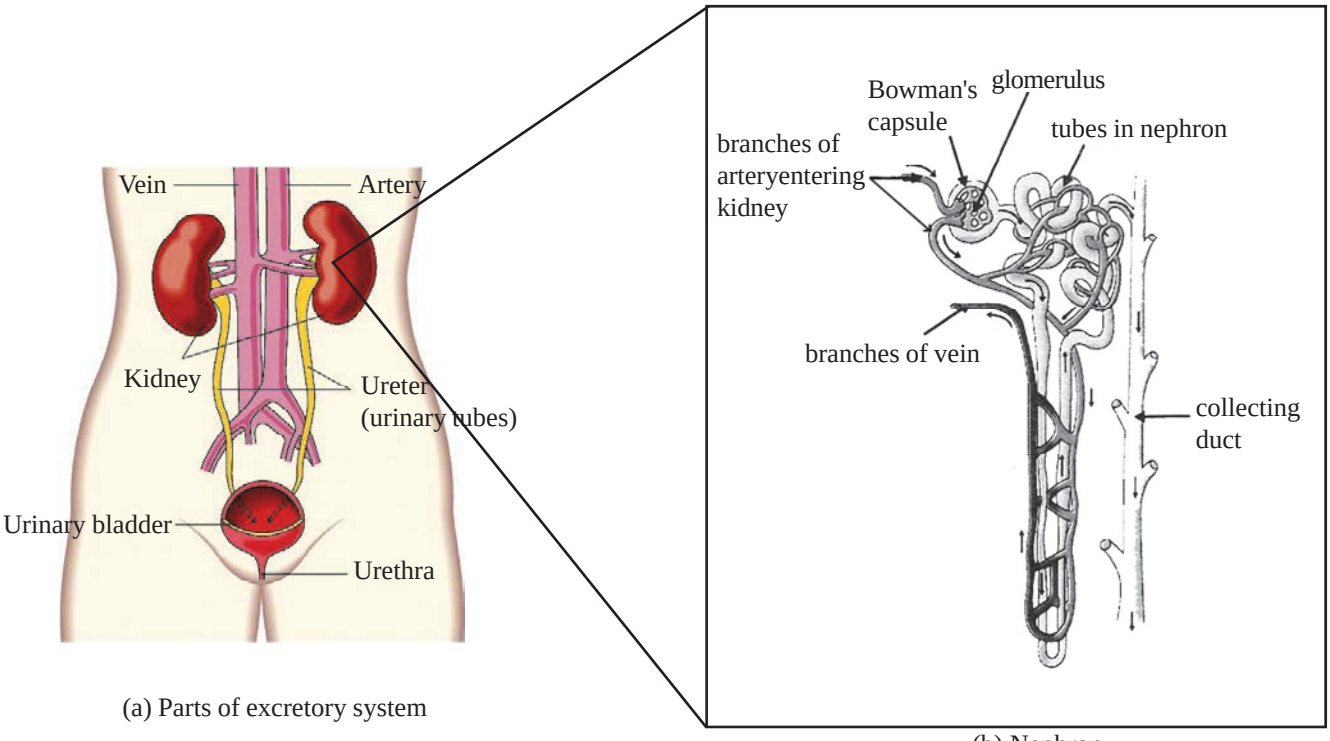
Several life processes go on in our body and several reactions either related to synthesis or breakdown takes place. Some products formed in them are utilized while others are expelled. Nitrogenous products like urea and uric acid formed by breakdown of proteins are expelled through our excretory system that mainly consist of a pair of kidneys.

- See figure 10A and write the names of the other major parts of this system

Kidneys are located at the back of body wall on both sides ofvertebral column. They are beanshaped, about the size of fist and dark reddish brown in colour. Blood passes to the heart from all parts of the body and from heart to different parts, kidney being one of them. Blood flows into our kidneys through arteries and flows out through veins. If we compare the substances present in the arteries and veins we would find a marked difference in their levels. For example the level of urea in arterial blood entering the kidney is very high ascompared to veinous blood leaving it.

Do you know?

About 170litres of blood flows through the kidneys of a human adult in a day. Only about 1.5 litres of urine is produced, while the rest of the fluid goes back into circulation.



(a) Parts of excretory system

(b) Nephron

Figure-11: Human excretory system

Structure and function of the unit of excretory system: the nephron

We have studied that gaseous exchange occurs across capillaries and alveolar walls in lungs. Similarly in our kidneys, exchange of substances occur across walls of the nephron and capillaries present over them. Some substances like urea, uric acid, salts, some proteins, water etc. flow out of blood vessels passing through the cup shaped structure called Bowman's capsule in the first part of a nephron. These flow through the tube of nephron. Several of these substances may enter back into the blood vessels around the nephron. The amount the would return to blood vessels depends on the level of these substances in the blood. The fluid that flows out of a nephron is urine which is collected via collecting ducts or tubes that collectively form a larger tube called the ureter.Urine passes through the ureter into the urinary bladder and gets collected there till adequate pressure created does not signal its expulsionoutside our body, through urethra. The muscles of the bladder are under control of nerves from our brain, thus the expulsion of urine may be controlled voluntarily to a certain extent.

Do you know?

Artificial kidneys are elaborate system of artificial tubes and membranes that help in removing excretory wastes. Often when our kidneys fail to function, the patients' blood is passed through this arrangement that removes wastes from the body. Blood from an artery is passed into the machine and after removal of waste, returned back to the body via a vein. The process of removal of waste is called dialysis which is normally carried out by our kidneys in our body

Several substances that are expelled from our body are useful for other organisms in the living world. We had studied earlier that nutrients cycle in nature and waste ofcertain organisms are nutrients of others.

- Which substances expelled by humans and other animals are useful for plants?
- How do plants use them?

Let us study life processes in plants to find out more about substances used by plants to synthesise other substances. We shall also study the processes in which plants synthesize certain substances, that makes the whole animal world dependent on them.

7.2 Life Processes in plants

People believed, for a very long time that life processes that occurred in animals did not occur in plants. Their notion was plants grew solely on nutients from the soil. This was even the belief of people like Aristotle nearly 2000 years ago. Life processes in plants started being studied extensively around 400 years back.

7.2.1 Nutrition in plants

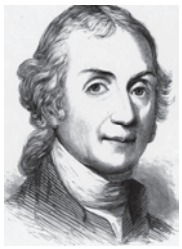


Von Helmont

A naturalist named as Von Helmont was studying plant nutrition and the role of water and soil in the 17th century. He took a 2Kg willow plant and planted it in a pot. He weighed soil that he added. Then he covered the pot with a perforated iron plate in such a manner that he could restrict the addition of extra soil or other materials to the pot. He watered the plant with measured quantity of water at regular intervals. He carried out the

experiment for a period of 5 years. After which he found that the plant weighed nearly 35 times while the soil in the pot decreased by only 50 grams. Thus he concluded that plants grew by nutrients from water but not from soil.

Thisexperiment not only broke the notion that plants derived all their nutrition from the soil, but also established the role of water in plant nutrition. Helmont had overlooked the role of air in the experiment. Actually, studies on composition of air and gases present in it had just started during his time. It took nearly another century to establish the role of gases in plant nutrition.



Joseph Priestley

A gas in which substances burnt vigorously was discovered by Joseph Priestley. Priestley had been experimenting on the effect of living and nonliving processes on air, effect of plants and animals etc. Figure- 11 illustrates some of his experiments.

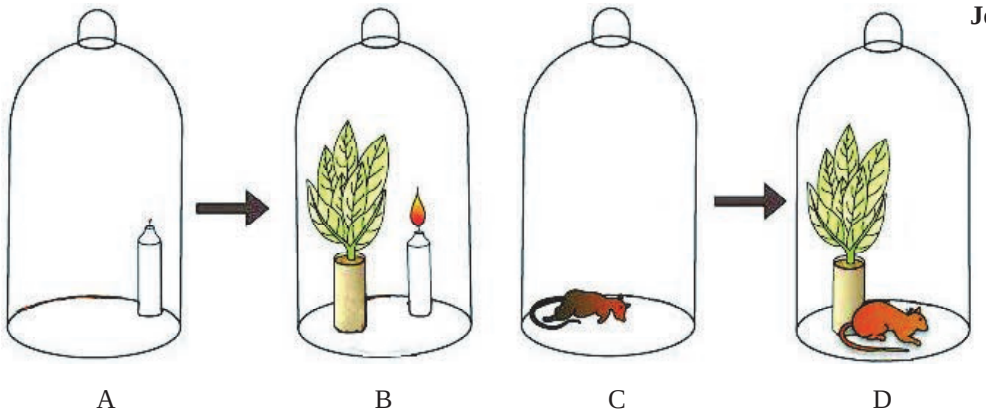


Figure-11: Priestley and his experiments

- What do you think Priestley may have observed in his experiments?
- How does the candle keep burning and rat remain alive in 'B'and 'D' respectively?

Priestley observed that the candle burnt out and the rat died in experimental set ups as in A and C respectively. The introduction of mint sprig in set ups as B and D kept the candle burning and the rat alive for a very long time. Priestley concluded that plants restored to air that part which was used up either by burning or by the process of breating in animals. Several years after this observation, this part of air which was a gas was named as 'Oxygen'.

People were relieved that as long as plants were present there would be ample oxygen for animals to survive. But the question was, what enabled plants to produce oxygen ?

We have studied that oxygen and glucose react to form CO2 and water and a large amount of energy is released. Scientists thought what would happen if CO2 and water were allowed to react?

It was observed that apart from respiration that occurred continuously during day and night, a process occurred simply during the day. During this, plants produced oxygen gas and carbohydrates. This process eventually depended on energy.

- Where did plants get this energy from?

J. Ingenhouz a scientist in the 18th century tried an experiment to answer this. Come let us also try doing this experiment.

Activity-8

You would require two beakers, two funnels, two test-tubes, Hydrilla (Cheela) /Elodea, black paper or cloth, a bucket full of water, incense sticks and match box.

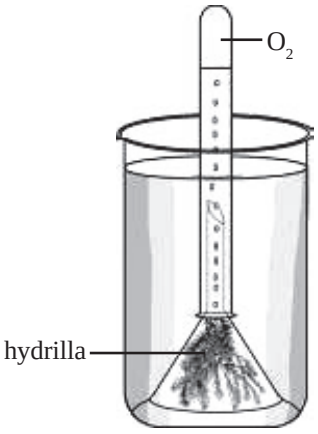


Figure-12: Apparatus with hydrilla

Arrange two sets of the experimental set up as shown in the figure12. The set ups have to be arranged under water (use the bucket full of water for the same). Ensure that the test-tube as well as funnel are full of water at the start of your experiment. Cover one set up with black paper and keep it in shade (ensure that it gets minimum heat and light) Let this be labeled as B. Keep the other set up, labeled as A in sunlight. Leave the set ups as such for a minimum of 3-4 hours. Keep observing from time to time. You would find bubbles rising up and gas collecting in the test-tube. Once the test tube of the set up A is around half filled with gas, take the whole set up under water and remove the test tube. Close its mouth with your thumb. Ensure that the gas does not escape. Now bring the test tube out of water, keep a burning incense stick ready and as soon as you remove your thumb, insert the incense stick into the test-tube.

- Did the incense stick start burning more brightly?
- Which gas do you think was collected in the test-tube(Remember testing a gas in earlier classes using a burning splinter or incense stick)?
- In which of the set ups, A or B was the collection of this gas more?

We found that more gas was collected in A. This showed that light was necessary for collection of the gas.The gas present was oxygen. Presence of which is tested by using a burning incense stick /a burning splinter. We found that sunlight was necessary for production of oxygen. This was what Ingenhouz had also concluded.

There is energy in sunlight and it was found that plants could use simple compounds to produce certain complex carbon compounds like pyruvate, glucose, starch, etc. in the presence of sunlight. They released oxygen in the process. Scientists call the process of production of complex substances from simple ones'synthesis'. Light dependent synthesis was thereby called photosynthesis (photo meaning light). Photosynthesis is the most important life process in nature. Most carbon compounds and the whole amount of oxygen in nature are produced by this process. All animals including humans essentially require these for their survival. Foods are mainly carbon compounds most of which may be synthesized from certain carbon compounds produced by photosynthesis. It is observed that the major carbon compound produced during photosynthesis is glucose and the major storage product is starch (produced mainly by combination of several glucose molecules).

- Water gives oxygen and hydrogen then where does carbon for carbon compounds come from?

The answer to this came from experiments done during the end of 18th century and beginning of 19th century. Scientists could state that a combination of same number of carbon and water molecules form carbon compounds and the source of carbon is the gas, carbon dioxide.

We know that as air flows in and out of the stomata and certain areas of loose tissue in the plant body. Carbon dioxide in the air flowing in is utilized by the cells carrying out photosynthesis.

Using the information so far, if we try to write an equation for photosynthesis it could be-

$nCO_2 + nH_2O \rightarrow Cn(H_2O)_n + nO_2$ 'n' represents number of molecules

- What will be the equation when the value of n is 6? What is the carbon compound formed?

We have studied about the equation of respiration in the previous sections of this chapter. If we compare both the equations of photosynthesis and respiration we shall find that they are reverse of each other. Carbon compounds and oxygen are products in photosynthesis, while they are reactants in respiration. Further, in the former, light energy is converted to chemical energy and synthesis of carbon compounds take place while in the latter, chemical energy is converted to other forms with the breakdown of glucose. This energy is utilized in running life processes.

We know that respiration occurs in all cells of a living body.

- Does photosynthesis also occur in all the cells of plants?

Research in this direction had started in the 19th century. It was observed that it occurred only in those cells that contained 'chloroplasts'. The green pigment in chloroplasts was named 'chlorophyll' about this time. After the invention of the electron microscope and advancement of staining techniques in the 20th century, the internal structure of chloroplast could be observed. The study of reaction sites was also possible. Scientists used staining techniques and observed that oxygen was produced by breakdown of water molecule only in the presence of light. Half the number of water molecules used in photosynthesis is expelled as products. Now, we may show the formation of glucose in photosynthesis by the following equation.

$$6CO_2 + 12H_2O \xrightarrow[\text{Sunlight (light energy)}]{\text{Chlorophyll}} C_6H_{12}O_6 + 6O_2 + 6H_2O$$

Let us observe chloroplasts.

Activity-9

We would require a Rheo leaf or algae, a compound microscope, slide, coverslip, needle/pin and watch glass. We would have to take out a thin layer from either the upper or lower surface of the leaf by tearing the leaf with a jerk. Then,prepare a slide of this and focus over a stomata. Now change over to higher magnification, set the microscope and observe the stomatal guard cells. Several green structures are observed. These are chloroplasts. To observe chloroplasts in algae, we would need a needle to separate out stands of algae. Take the finest stand, prepare a slide and observe under microscope.

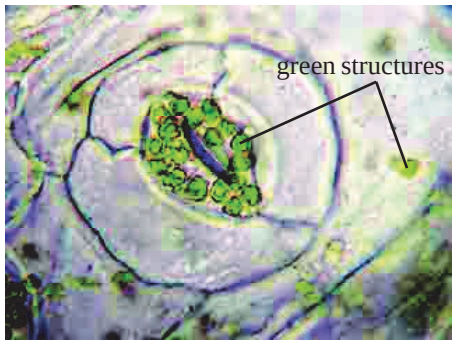


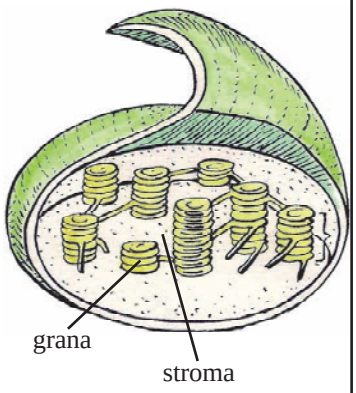
Figure-14: stomata and chloroplasts in Rheo leaf

The green structures in the strands are chloroplasts. Make a sketch of the observation.

- What substances do plants require for photosynthesis?
- How do these substances reach the chloroplasts?

Do you know?

Chloroplasts are twice the size of mitochondria. They are structures enclosed in double membranes. There is a third membrane that folds into stacks of sack like structures called as thylakoids. The stacks of sacks (that resemble stacks of coin) are called as grana and the fluid around these is called as stroma. Julius Von Sachs, a 19th century scientist had for the first time shown the presence of chloroplasts in which the photosynthetic pigment chlorophyll was present. He could not isolate chloroplasts but could further suggest that starch granules are formed in green leaves as a result of photosynthesis. It was later observed that light dependent processes of photosynthesis occurred in the grana where oxygen was released by the breakdown of water molecule. The reactions that lead to synthesis of carbon compounds occurred in the stroma and did not require the presence of light. Apart from sugars, amino acids, fatty acids and other compounds are synthesized in the chloroplasts. Chloroplasts also play a very important role in immune system of plants.



We had studied in class 9 that chloroplasts and mitochondria are examples of organelles formed by endosymbiosis. These cannot be manufactured by the cell, rather the genetic material of these pass on from one generation of cells to the other ensuring their development in each further generation. The seeds of plants have genetic material of chloroplasts.

7.2.2 Structure and function of transport systems in plants

We have studied that substances required for life processes and their products(including the excretory products) are transported across the body in elaborate system of tubes filled with fluids. Plants also have elaborate system of tubes for transport through which different types of substances are transported. We had studied about the vascular tissues xylem and phloem in plants last year. The pattern of arrangement of these may be observed in several plants. Flowering plants usually have patterns of arrangement as shown in Figure- 15 A and B. We usually find vascular tissue located centrally in the roots while in stems they are usually located towards the periphery. These are usually found arranged in concentric circles in dicotyledenous flowering plants. We find vascular tissues scattered in monocotyledenous flowering plants(See Figure-15 A).

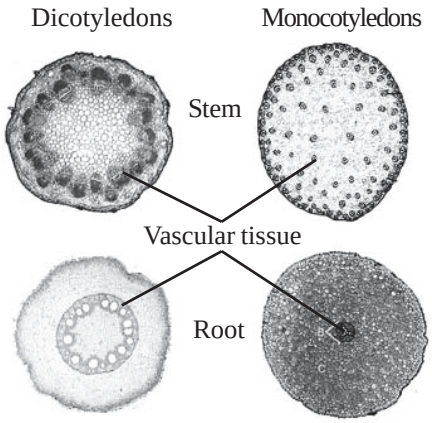


Figure-15(a): vascular tissues scattered in monocotyledenous flowering plants

Apart from stems and roots vascular tissues are also found in parts like theflower, leaf, fruits etc. Several substances like salts and nitrogenous compounds dissolved in water are transported through xylem from roots to other parts of the plant body. Substances are transported through xylem in several ways. The amount of water in soil being greater than that in root cells allows diffusion of water into root cells and to the xylem. The transpiration loss due to movement of water out of the stomata generates vaccum in the xylem tubes reaching the stomata, called as traspiration pull. This facilitates upward movement of water through the xylem. This helps in lifting water to great distances in plants. Most cells of phloem are living cells, thus movement of substances like certain sugars, amino acids etc. through phloem is often mediated by these cells at the expense of energy. Often when flow is towards gravity it hardly requires any energy.

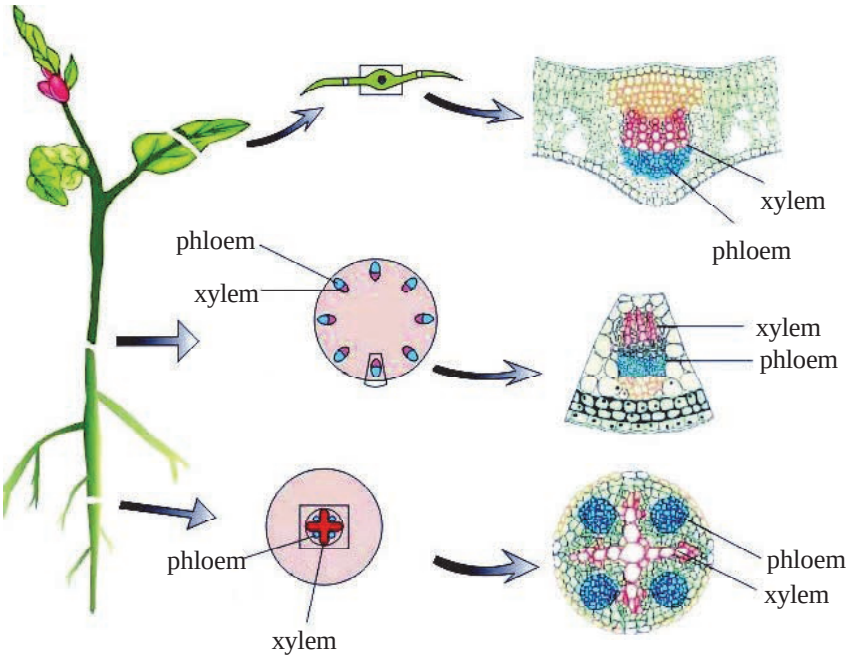


Figure-15(b): Location of vascular tissue in different parts of a dicotyledenous plant

7.2.3 Removal of wastes in plants

Most of the substances in plants are utilized in some or the other manner. The waste of one process being used in the other. Take for example, CO2. This gas is expelled from the body of animals while in plants, most of it is utilized in the process of photosynthesis during day time. It is expelled from the plant body only at night. On the contrary excess O2 (exceeding the amount used in respiration) produced during daytime is expelled from the plant body.

Transpiration (mainly a process of evaporation) causes loss of water and some salts from different parts of plant. Most carbon compounds that are wastes are usually stored away in dead tissues of the plant body.These are removed via leaves, flowers and bark that fall off from time to time. Certain salts are also stored in dead tissues. They are usually present in cell vacuoles. These are often expelled out of the plant body. In water plants they are directly released into the water.

Do you know?

Several substances usually called as excretory products, like latex, gums, resins, tannins, alkaloids etc. These usually have protective functions for the plant. We often use many of these like the latex of rubber plant is used to produce rubber.

Mangroves are a group of plants that grow in salty marshes. The excess salt is expelled through salt glands present at the base of the leaves of such plants.

7.3 Life processes in unicellular organisms

Figure-16 shows intake of food in Amoeba. We have studied that the bodies of unicellular organisms are in direct contact with the environment, thus gaseous exchange, food and excretory products move directly across the cell membrane or any specific part of it. Life processes of these organisms are carried out by different parts of the cell.

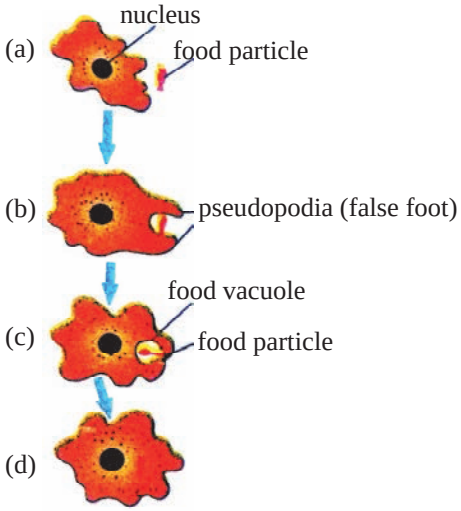


Figure-16: Intake of food in Amoeba

Keywords

Enzymes, peristalsis, villi, alveoli, double circulation, photosynthesis, adenosine tri phosphate,



What we have learnt

1. Human digestive system comprises of digestive tube and glands.
2. The main parts of digestive tube are mouth, pharynx, oesophagus, stomach,small intestine, large intestine and anus
3. Food is oxidised in the presence of oxygen , the products being water and carbon dioxide. A large amount of energy is released in the process which is stored as ATP (adenosine tri phosphate) to be used for life processes.
4. ATP is called as energy currency.
5. Human excretory system comprises of a pair of kidneys, a pair of ureter, a urinary bladder and urethra.Kidneys are composed of several nephrons.

6. Human transport system is mainly a blood vascular system having heart and blood vessels(arteries and veins)
7. Arteries carry blood away from the heart while veins carry blood towards the heart.
8. Plants are producers as they synthesize carbon compounds like sugars, starch etc. in the presence of sunlight (or any light of similar intensity) with the help of chlorophyll, by the process of photosynthesis. Most of the living world depends on plants.
9. Photosynthesis occurs in chloroplasts.
10. Plants convert light energy to chemical energy .
11. Xylem and phloem are mainly vascular tissues of plants that are present in all the parts of plants(stem, root, leaf, fruit, flower etc.)
12. Some excretory products are expelled from the plant via falling leaves, flowers etc. Some others are stored in vacuoles and expelled from the plant body from time to time.
13. Gaseous exchange occurs mainly through stomata and loose tissues in plants.

Exercise

1. Choose the right option
   (i) The internal wall of the stomach is usually not injured by HCl because of-
   (a) Pepsin (b) mucous (c) salivary amylase (d) none of these
   (ii) Gaseous exchange occurs during respiration between-
   (a) respiratory and digestive tract (b) alveoli of lungs and surrounding capillaries
   (c) air alveoli and respiratory tubes (d) respiratory tube and pharynx
   (iii) Oxygenated blood flows from lungs to heart by-
   (a) pulmonary artery (b) pulmonary vein
   (c) pharynx (d) none of these
   (iv) Kidney is composed of several-
   (a) nephron (b) Bowman's capsule
   (c) ureter (d) urinary bladder
2. How is a vein different from an artery?
3. Differentiate between respiration in the presence and absence of oxygen.
4. What are the differences between the sources of energy of producers and consumer? Clarify this.
5. What is the mutual relationship of animals and plants on the basis of food?
6. Why is formation of urine and its removal necessary for plants?
7. What is the difference between substances transprted by xylem and phloem?
8. What is the functionof HCl secreted in the stomach?

9. Cover some branches of a plant with polythene and tie it up. After about 3-4 hours measure the volume of water collected in it. Can you give an estimate of the total water loss by the plant? Justify your answer.

10. Several experiments on transpiration are conducted in agriculture. One such example is given here, 4 leaves of an arhar plant were coated with vaseline. Observe the table and answer the following questions-

Table: Reduction of weight of leaves due to transpiration

Leaf number	Surface on which vaseline was applied	Loss of weight
1	None of the surfaces	40%
2	Both surfaces	2%
3	Upper surface	36%
4	Lower surface	4%

- Maximum transpiration took place from which surface of the leaf?
- Which surface do you think has the maximum number of stomata?
- What happens when a large amount of water is lost from a plant?
- How will a plant be affected if water fails to move out of it?

11. Explain how blood flows in two pathways from the heart?

Annexure

Test for fats

Rub or put a small amount of the substance to be tested on a piece of white paper. Allow it to dry. If the paper turns oily and transluscent, then the substance tested contains fat.

Test for proteins

Take 10 drops of substance to be tested in a test tube. If the substance is a solid, grind it and put a pinch of it in the test tube and add 10 drops of water to it. Now add 2 drops of 2% copper sulphate and 10 drops of 10% caustic soda solution.

If the colour turns violet or purple, then the substance contains protein.

Test for carbohydrates

Prepare solution of substance to be tested, add 2-4 drops of dilute iodine solution(1%). A blue black colour gives conformatory test for starch. Add iodine directly over a cut potato for test.